APPLIED NETWORKING LAB

Enterprise Network Design & Security Hardening

Segmented VLAN Infrastructure with Layer 2/3 Threat Defense — Designed, Deployed, and Verified in Cisco Packet Tracer

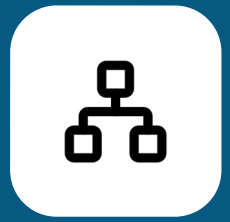
VLAN DHCP NAT EtherChannel Port Security DHCP Snooping DAI SSH



Cisco Packet Tracer • CCNA Practical Project • 2026

01

Project Overview



A small-enterprise LAN, built and defended end to end.

This project simulates a small organization's network across two connected phases: first building a segmented, routed, and redundant LAN — then hardening it against real Layer 2/3 attacks.

Segmentation

Three departments — HR, IT, Sales — isolated on dedicated VLANs across two access switches.

Routing & Redundancy

Router-on-a-stick inter-VLAN routing with EtherChannel-bundled switch links for resilience.

Threat Defense

Port Security, DHCP Snooping, Dynamic ARP Inspection, VLAN ACLs, and SSH-only management.

Live Troubleshooting

A real DHCP fault was diagnosed and resolved using a layer-by-layer, evidence-based approach.

02

Project Objectives

VLAN Segmentation

Create and assign HR, IT, and Sales VLANs across two access switches.

Inter-VLAN Routing

Configure router-on-a-stick so departments can reach each other through R1.

Automatic Addressing

Deploy per-VLAN DHCP pools so every host is addressed without manual entry.

Redundant Switching

Bundle switch-to-switch links with EtherChannel (LACP) for failover and throughput.

Internet Access

Enable NAT overload so all VLANs share a single outside address to reach the internet.

Endpoint & Access Control

Apply Port Security, VLAN ACLs, and SSH-only management to close common attack paths.

Spoofing & Storm Defense

Enable DHCP Snooping, Dynamic ARP Inspection, and broadcast storm control.

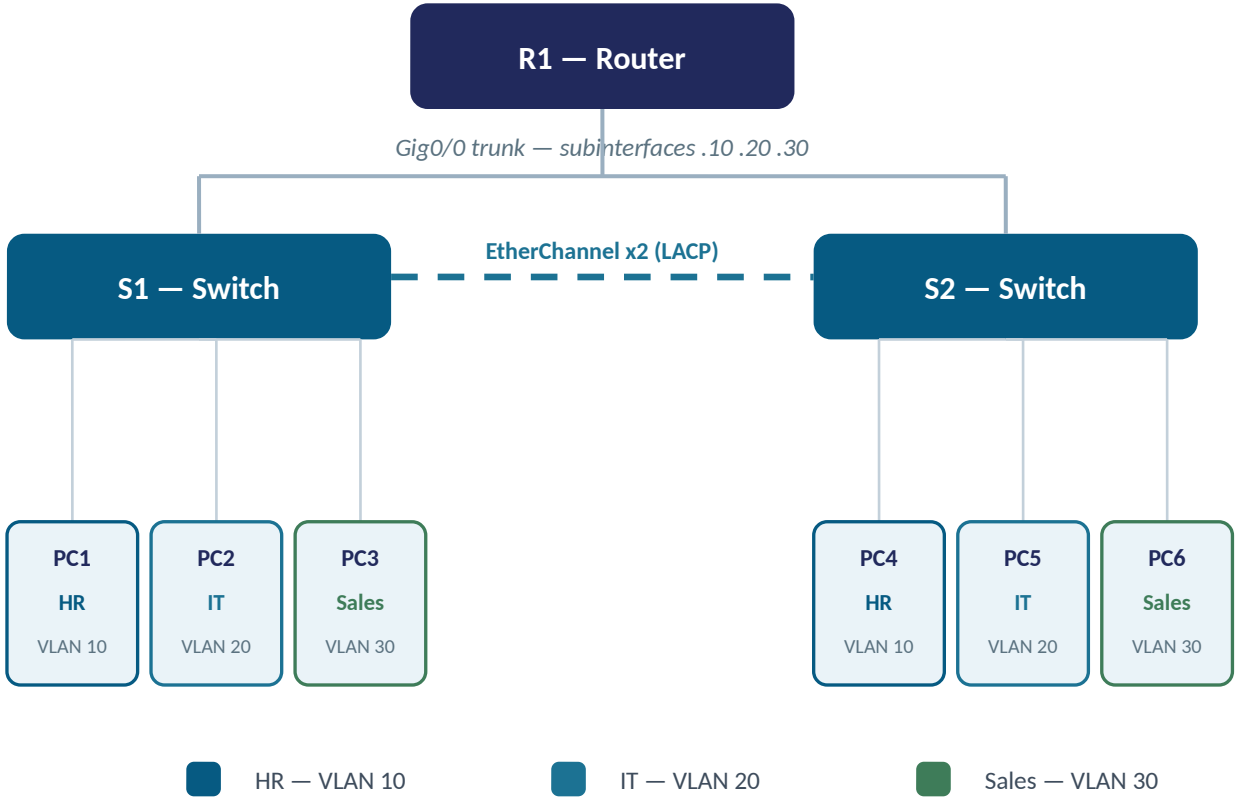
Validated Fault Response

Simulate an attacker and a live fault, then trace and resolve each with CLI evidence.

03

Network Topology

Router-on-a-stick core, redundant access layer, six segmented hosts



192.168.10.0/24 • 192.168.20.0/24 • 192.168.30.0/24 — gateways .1 on each VLAN subinterface

04

Phase 1 — Core Infrastructure

VLAN Segmentation

`vlan 10/20/30`

HR, IT, and Sales isolated into separate broadcast domains on both access switches.

Router-on-a-Stick

`gig0/0.10 / .20 / .30`

Three 802.1Q subinterfaces on R1 route between VLANs over a single trunk link.

DHCP Automation

`ip dhcp pool`

Per-VLAN pools auto-assign address, gateway, and DNS — no static configuration.

EtherChannel (LACP)

`channel-group 1 mode active`

Two S1–S2 links bundled into one logical, load-balanced, fault-tolerant path.

NAT Overload

`ip nat inside source ... overload`

All three VLANs share a single outside address for outbound internet access.

Verified End-to-End

`show / ping / trace`

Every stage confirmed with IOS show commands before moving to the next.

05

Case Study — Live Fault Diagnosis

Hosts received APIPA addresses instead of DHCP leases — traced from symptom to root cause using CLI evidence, not guesswork.

Step 1

Symptom

PCs showed 169.254.x.x (APIPA) — no DHCP lease obtained on any VLAN.

Step 2

Check Router

show ip dhcp binding returned no entries — request never reached R1.

Step 3

Check Switch

show interfaces trunk on S1 returned empty — no port was trunking.

Step 4

Isolate Port

show cdp neighbors confirmed the exact physical link to R1: FastEthernet0/5.

Step 5

Apply Fix

switchport mode trunk + allowed vlan all on Fa0/5; PortFast added to access ports.

Step 6

Verified

PCs re-leased correctly; ping and inter-VLAN routing confirmed working end-to-end.



PHASE 2

Security Hardening

Simulating an insider threat and defending the network layer by layer — endpoint access, DHCP integrity, ARP trust, and encrypted management.

06

Security Controls Implemented

Port Security

One MAC per port, sticky-learned, violation = shutdown. Blocks any unauthorized device plugged into the LAN.

VLAN Access Control

Router ACL denies HR → IT traffic while permitting all legitimate inter-VLAN paths.

DHCP Snooping

Only the router-facing trunk is trusted; rogue DHCP offers from any other port are dropped.

Dynamic ARP Inspection

Validates ARP packets against the DHCP snooping table to block spoofing / MITM attempts.

Storm Control

Broadcast traffic capped at 10% of port bandwidth to contain loops and flood-based DoS.

SSH-Only Management

RSA keys + local auth; Telnet disabled so remote administration is fully encrypted.

07

Attack Simulation & Defense Outcomes

An attacker host was placed on the LAN and three attack vectors were tested against the configured defenses.

Attack Vector	Defense Mechanism	Verification Command	Outcome
Unauthorized device plugged into an access port	Port Security (max 1, sticky, violation shutdown)	`show port-security`	**Port shut down**
Rogue DHCP server offering fake gateway	DHCP Snooping (untrusted ports)	`show ip dhcp snooping`	**Offer dropped**
ARP spoofing attempt (MITM)	Dynamic ARP Inspection	`show ip arp inspection`	**Packet denied**
HR host attempting to reach IT VLAN	VLAN Access Control List	`show access-lists`	**Traffic denied**
Broadcast flood on an access port	Storm Control (10% threshold)	`show storm-control broadcast`	**Traffic capped**
Telnet-based remote access attempt	SSH-only VTY lines	`show ip ssh`	**Telnet refused**

08

Skills Demonstrated

Networking

- VLAN design & trunking
- Router-on-a-stick routing
- DHCP & NAT configuration

Redundancy

- EtherChannel (LACP)
- Spanning Tree behaviour
- PortFast tuning

Security

- Port Security
- DHCP Snooping & DAI
- VLAN ACLs, SSH hardening

Troubleshooting

- CLI-driven fault isolation
- CDP-based topology checks
- Evidence-based root cause

Tools: Cisco Packet Tracer • Cisco IOS CLI



Project Completed & Verified

A segmented enterprise LAN — designed, automated, made redundant, connected to the internet, and hardened against real attack vectors.

10

Verification Tests Passed

6

Security Controls Deployed

1

Live Fault Diagnosed & Resolved

ENTERPRISE VLAN SEGMENTATION & INTER-VLAN ROUTING INFRASTRUCTURE

VLANs, Router-on-a-Stick, DHCP, EtherChannel, NAT & Troubleshooting

CCNA / Cyber Security — Practical Networking Project

Name: __Ahmad Hassan_____

Course: Diploma in Cyber Security

Module: CCNA / Computer Networking

Lab: CCNA Mega Lab 1

Software: Cisco Packet Tracer

Title: Segmented Enterprise LAN Deployment: From Design to Root-Cause Troubleshooting

Date: 24-09-2026

Project Type : _Netwroking Engenaring_____

1. PROJECT OVERVIEW

This project presents the design, configuration, verification, and troubleshooting of a small-enterprise segmented LAN using Cisco Packet Tracer. The simulated organization consists of three departments — HR, IT, and Sales — each isolated on its own VLAN and distributed across two access switches.

The network implements inter-VLAN routing using a router-on-a-stick design, automatic IP assignment through DHCP, redundant switch-to-switch connectivity via EtherChannel, and internet access through NAT overload.

The main purpose of the project is not only to configure the network, but also to demonstrate a professional network-engineering workflow:

Design → VLAN Segmentation → Trunking → Router-on-a-Stick → DHCP → EtherChannel → NAT → Verification → Fault Isolation → Troubleshooting → Final Validation

The lab specifically includes VLAN creation, 802.1Q trunking, DHCP pool configuration, EtherChannel (LACP) redundancy, NAT overload for internet access, and hands-on troubleshooting of a real DHCP failure encountered during implementation.

2. PROJECT OBJECTIVES

1. To design a segmented, multi-VLAN enterprise topology in Cisco Packet Tracer.
2. To create and configure VLANs for HR, IT, and Sales departments.
3. To configure 802.1Q trunk links between switches and the router.
4. To implement router-on-a-stick for inter-VLAN routing.
5. To configure DHCP pools for automatic IP assignment per VLAN.
6. To bundle redundant switch-to-switch links using EtherChannel (LACP).
7. To configure NAT overload for internet access from all VLANs.
8. To verify VLAN, trunk, DHCP, and EtherChannel operation using Cisco IOS show commands.

9. To perform end-to-end connectivity testing between VLANs.
10. To diagnose and resolve a real DHCP failure caused by a missing trunk configuration.

3. NETWORK SCENARIO

The organization has three departments distributed across two access switches:

- HR (VLAN 10)
- IT (VLAN 20)
- Sales (VLAN 30)

The network must support:

- Logical segmentation of department traffic
- Inter-VLAN communication via a central router
- Automatic IP addressing (DHCP)
- Redundant switch-to-switch links (EtherChannel)
- Internet access for all VLANs (NAT)
- Fault isolation and troubleshooting

R1 acts as the central router performing inter-VLAN routing, DHCP service, and NAT. S1 and S2 are access switches, each hosting one PC per VLAN.

4. NETWORK DEVICES

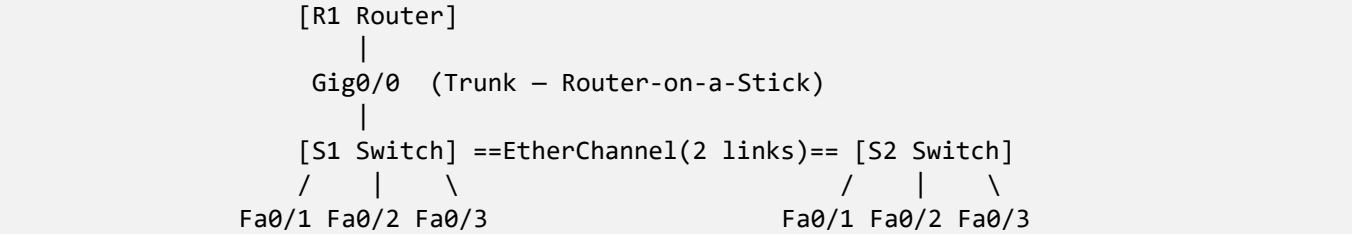
Device	Quantity	Purpose
Cisco 1941 Router	1	Inter-VLAN routing / DHCP / NAT
Cisco 2960 Switch	2	VLAN access switching
End Devices / PCs	6	Connectivity & DHCP testing
Cisco Packet Tracer	1	Network simulation

VLAN & Port Distribution

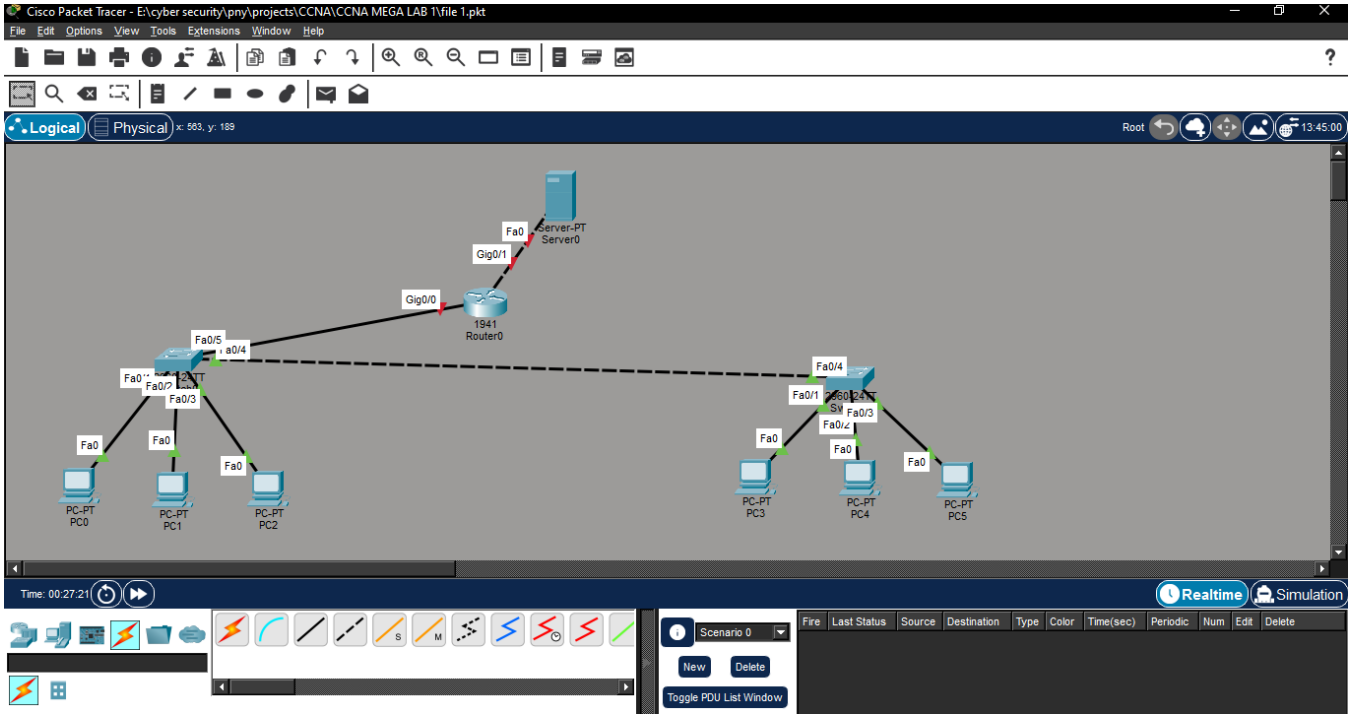
Switch	Port	VLAN	PC
S1	Fa0/1	10 (HR)	PC1
S1	Fa0/2	20 (IT)	PC2
S1	Fa0/3	30 (Sales)	PC3
S2	Fa0/1	10 (HR)	PC4
S2	Fa0/2	20 (IT)	PC5
S2	Fa0/3	30 (Sales)	PC6

5. NETWORK TOPOLOGY

The topology consists of a central router (R1) connected to Switch S1 via a trunk link, S1 and S2 interconnected via a redundant EtherChannel bundle, and three PCs on each switch representing HR, IT, and Sales.



```
          |     |     |                       |     |     |
        [PC1] [PC2] [PC3]                   [PC4] [PC5] [PC6]
        (HR)  (IT) (Sales)                  (HR)  (IT) (Sales)
```



6. IP ADDRESSING PLAN

Department	VLAN	Network	Default Gateway
HR	10	192.168.10.0/24	192.168.10.1
IT	20	192.168.20.0/24	192.168.20.1
Sales	30	192.168.30.0/24	192.168.30.1

All host addressing within each VLAN is assigned dynamically via DHCP, with the gateway address excluded from each pool.

7. PHASE 1 — VLAN CREATION (S1 & S2)

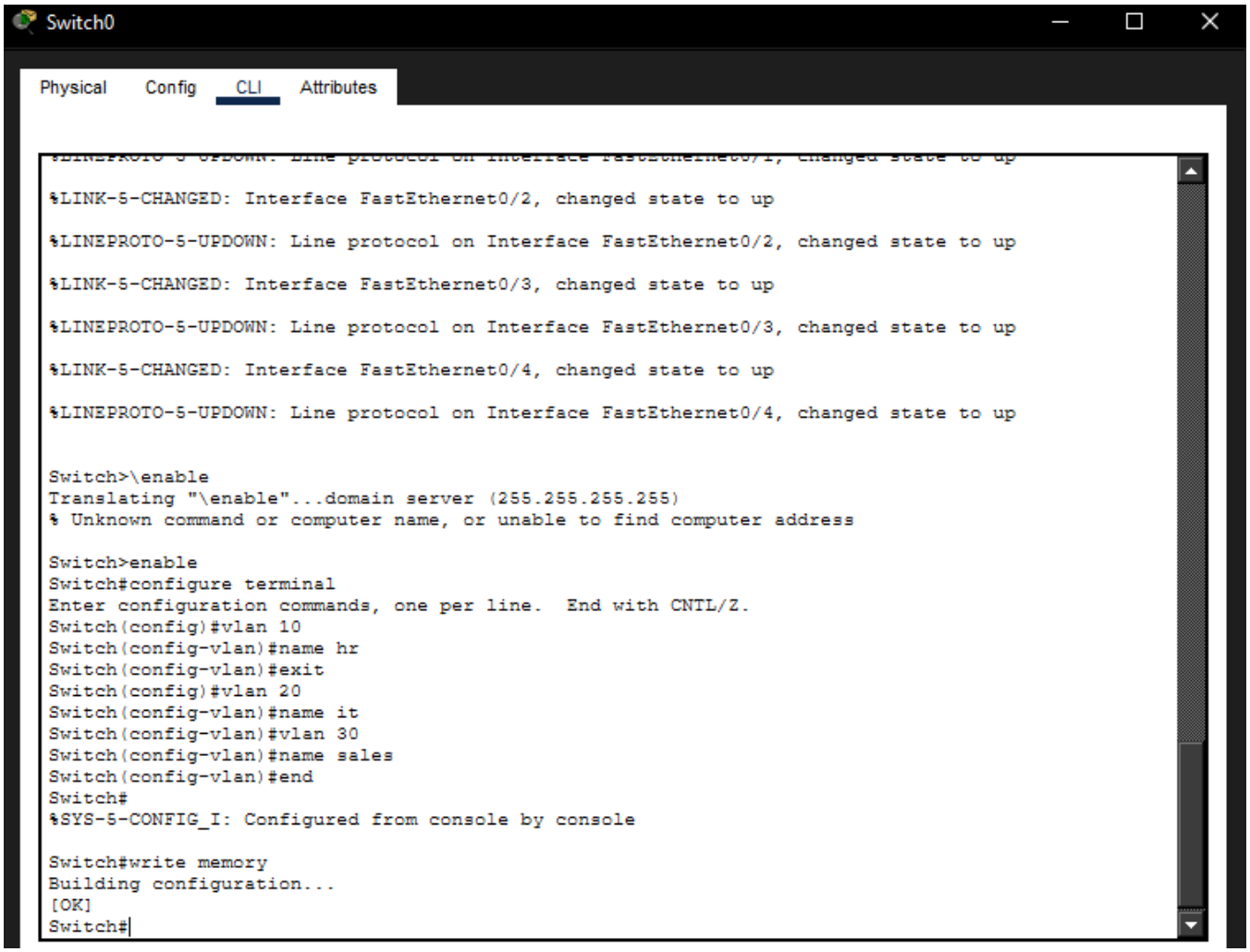
VLANs 10 (HR), 20 (IT), and 30 (Sales) were created identically on both access switches, and each access port was assigned to its department VLAN.

```
enable
configure terminal
vlan 10
 name HR
vlan 20
 name IT
vlan 30
 name SALES
exit
interface fa0/1
 switchport mode access
 switchport access vlan 10
```

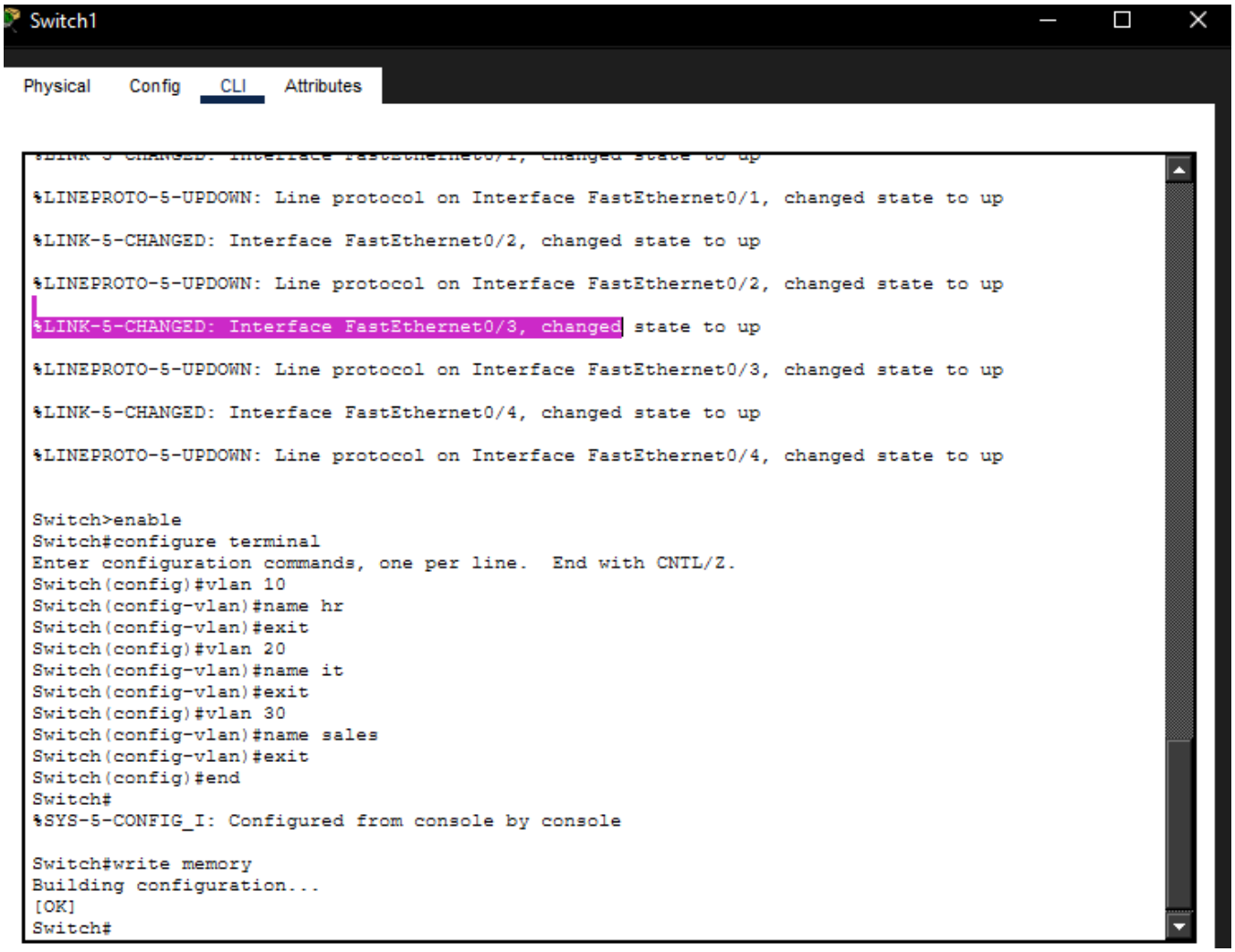
```
interface fa0/2
 switchport mode access
 switchport access vlan 20
interface fa0/3
 switchport mode access
 switchport access vlan 30
```

Verification:

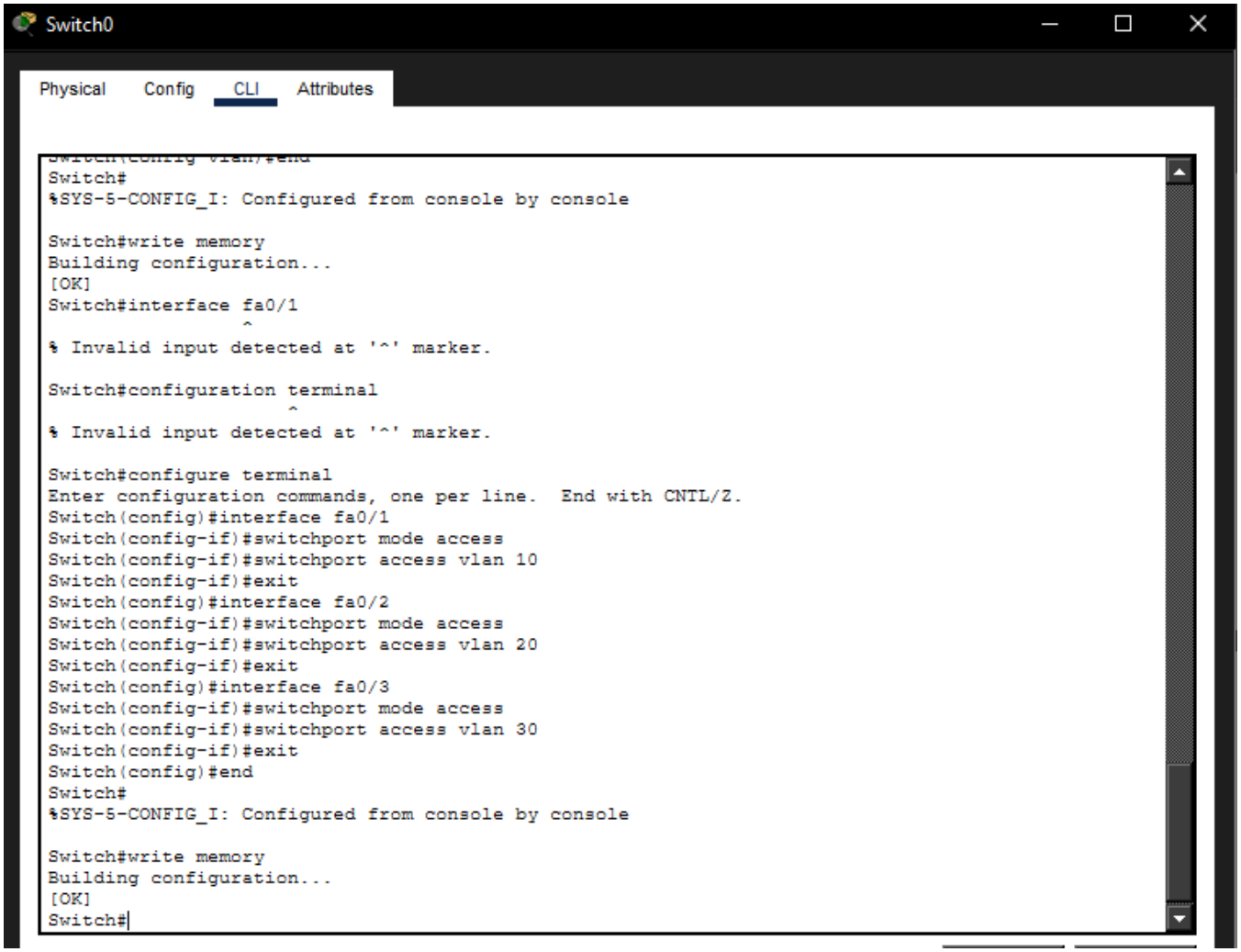
```
show vlan brief
```



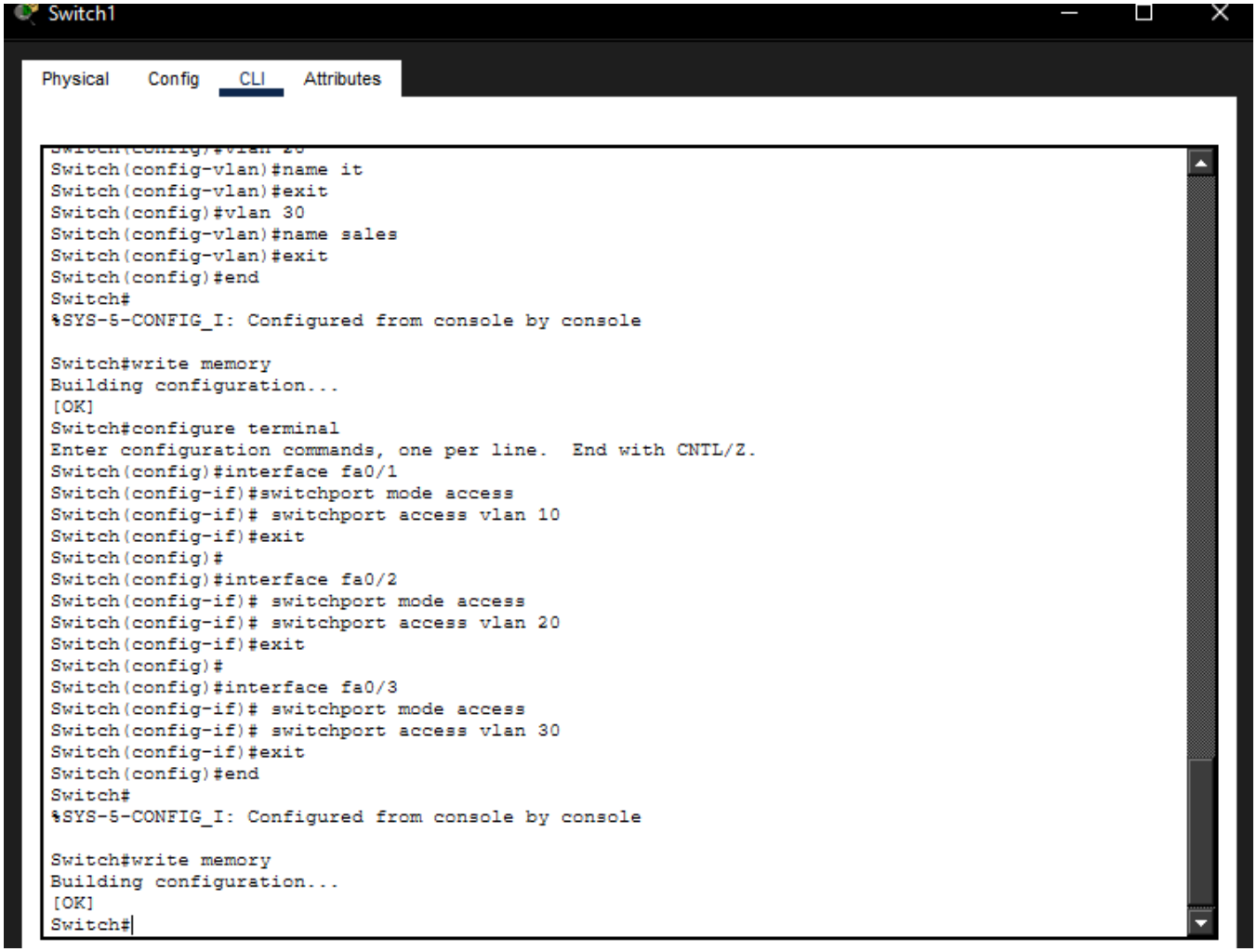
Vlan configuration SW1



Vlan configuration SW2



Assigning ports on SW1



Assigning ports on SW2

8. PHASE 2 — TRUNKING & ETHERCHANNEL (S1 ↔ S2, S1 ↔ R1)

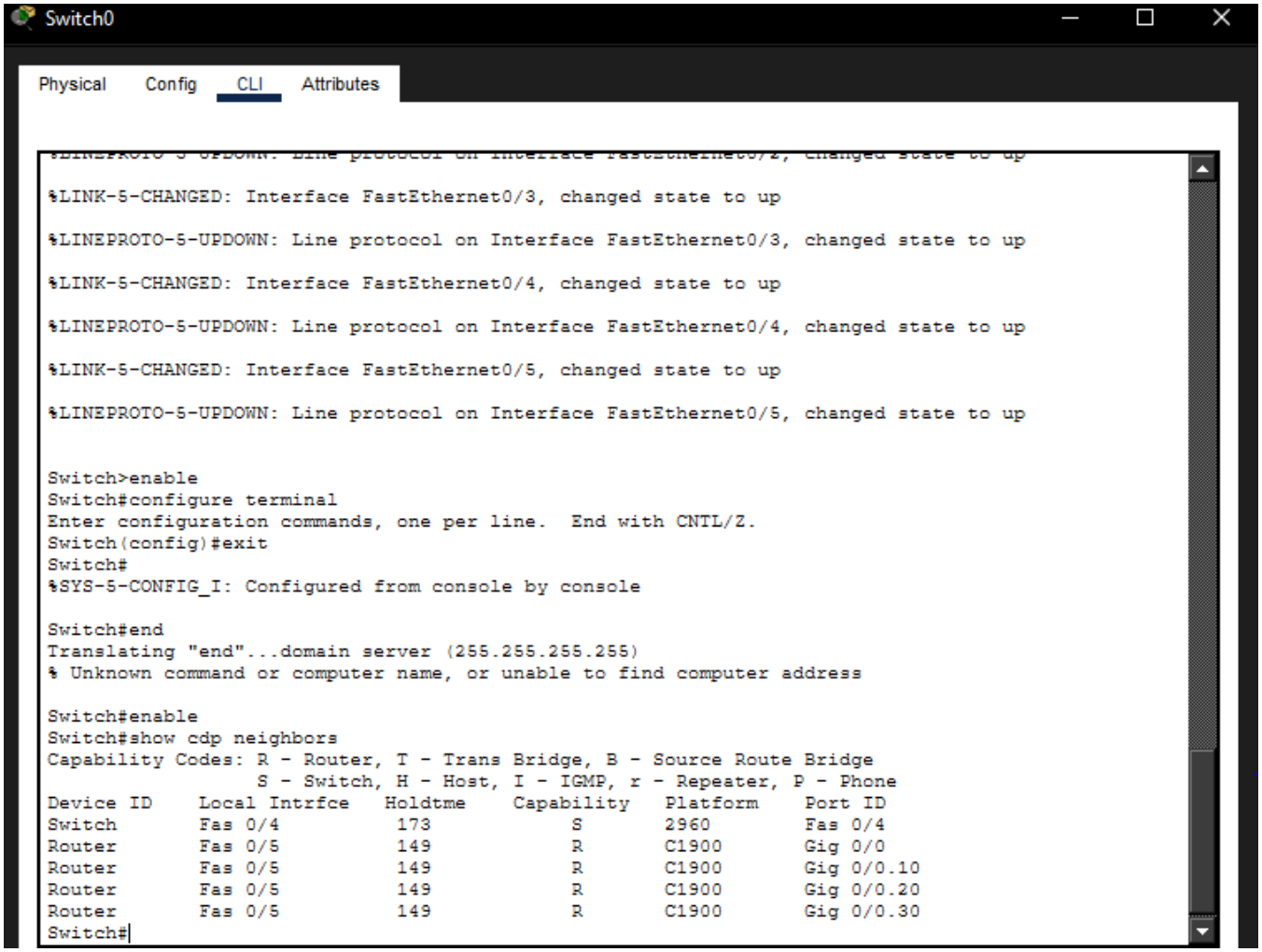
The two S1–S2 links were bundled into a single logical EtherChannel using LACP for redundancy and increased throughput.

```
interface range fa0/23 - 24
 switchport mode trunk
 channel-group 1 mode active
interface port-channel 1
 switchport mode trunk
```

The router-facing port on S1 was identified using CDP and set to trunk mode:

```
show cdp neighbors
interface fastEthernet0/5
 switchport mode trunk
 switchport trunk allowed vlan all
```

Why: trunking allows all three VLANs to traverse a single physical link; EtherChannel prevents STP from blocking one of the redundant S1–S2 links while combining their bandwidth.



CDP Neighbor Discovery

9. PHASE 3 — ROUTER-ON-A-STICK (R1)

R1's physical interface was subdivided into three 802.1Q subinterfaces, one per VLAN, each serving as that VLAN's default gateway.

```
interface gig0/0
 no shutdown
interface gig0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface gig0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
interface gig0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
```

Verification:

```
show ip interface brief
```

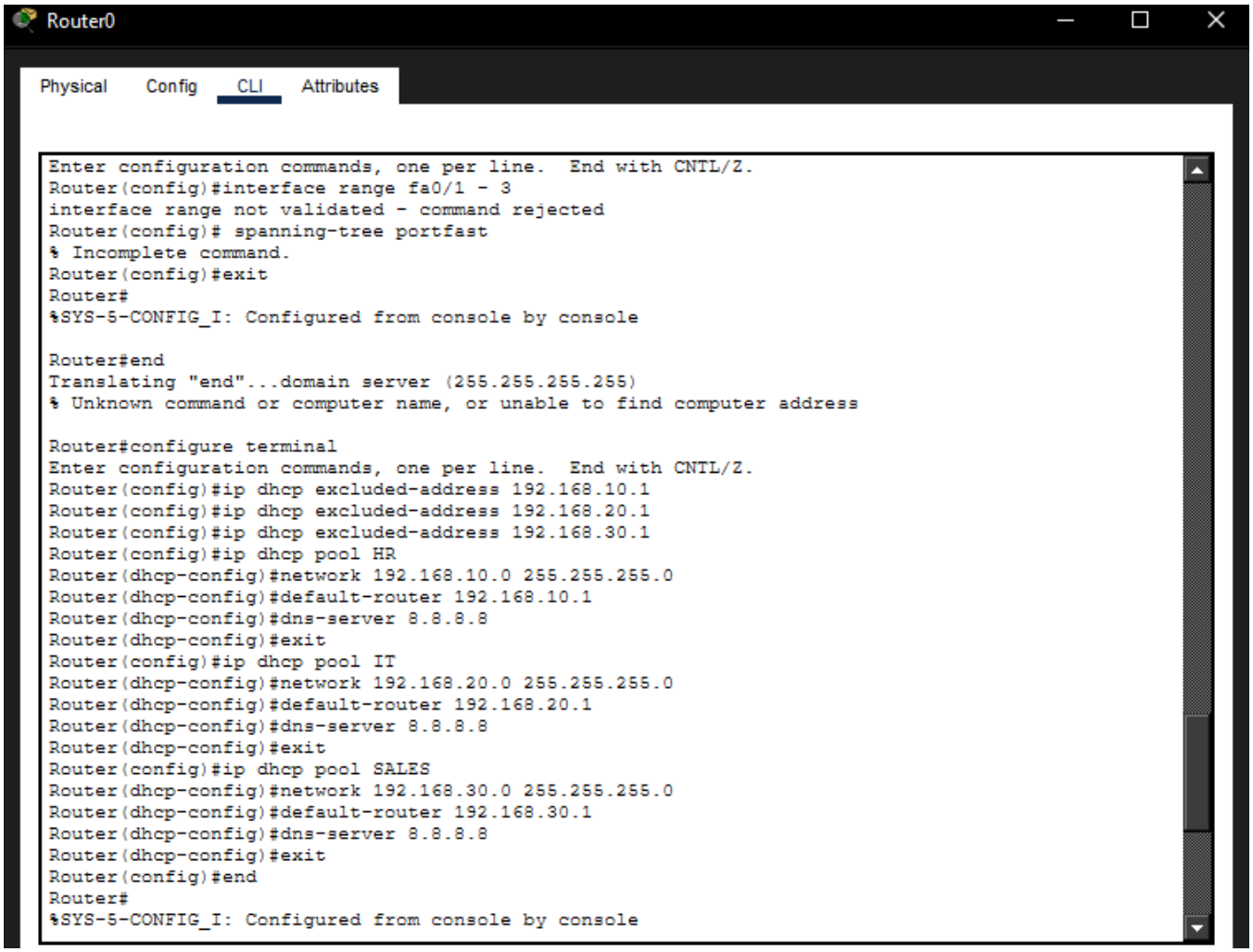


R1 Subinterface Configuration Verified via show ip interface brief (all links up/up)

10. PHASE 4 — DHCP CONFIGURATION (R1)

DHCP pools were created for each VLAN so hosts receive IP addressing, gateway, and DNS server automatically.

```
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.30.1
ip dhcp pool HR
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool IT
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
ip dhcp pool SALES
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
```



11. PHASE 5 — NAT OVERLOAD (INTERNET ACCESS)

NAT overload was configured so all three internal VLANs share a single outside IP address for internet access.

```
interface gig0/0.10
 ip nat inside
interface gig0/0.20
 ip nat inside
interface gig0/0.30
 ip nat inside
interface gig0/1
 no shutdown
 ip nat outside
access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface gig0/1 overload
ip route 0.0.0.0 0.0.0.0 200.1.1.1
```

Verification:

```
show ip nat translations
show ip nat statistics
```



12. TROUBLESHOOTING — DHCP FAILURE (REAL FAULT ENCOUNTERED)

During implementation, PCs failed to obtain a DHCP-assigned address and reported an APIPA (169.254.x.x) address, indicating the DHCP request never reached R1.

Diagnostic sequence followed:

1. show ip dhcp binding on R1 — returned no entries, confirming no DHCP request had arrived at the router.
2. show interfaces trunk on S1 — returned an empty result, revealing that no port on the switch was operating in trunk mode.
3. show cdp neighbors on S1 — used to positively identify the physical port connected to R1 (FastEthernet0/5), rather than assuming it.
4. The router-facing port was found to still be in default access mode, so 802.1Q VLAN tags were being dropped before reaching the router.

Resolution:

FastEthernet0/5 on S1 was explicitly set to trunk mode with all VLANs permitted; spanning-tree PortFast was also applied to the PC-facing access ports so they transition directly to forwarding state instead of waiting through STP's listening/learning delay.

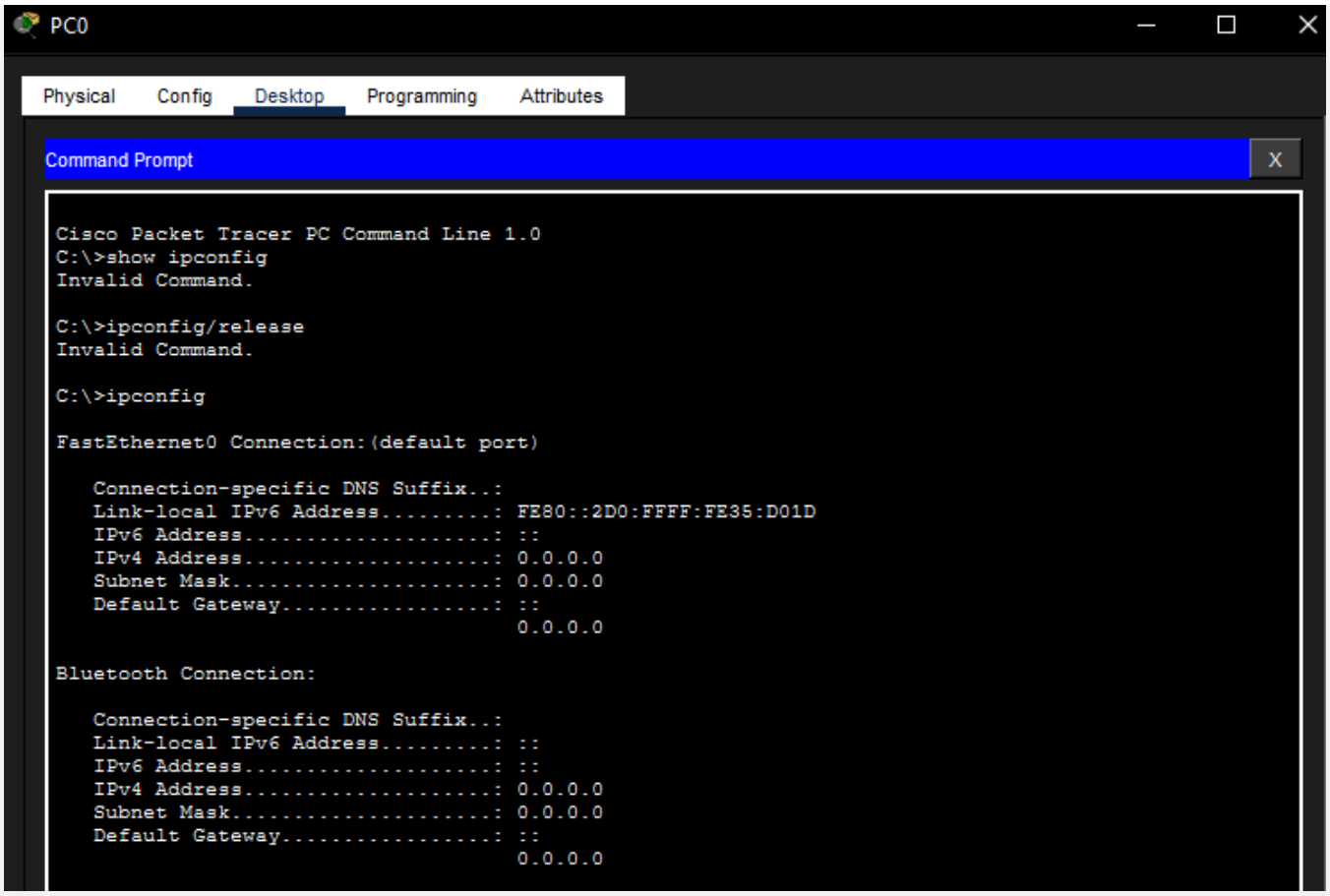
```
interface fastEthernet0/5
 switchport mode trunk
 switchport trunk allowed vlan all
```

```
interface range fastEthernet0/1 - 3
 switchport mode access
 spanning-tree portfast
```

After applying the fix, PCs were toggled from Static back to DHCP in IP Configuration to trigger a new lease request. Devices then successfully obtained addresses within their correct VLAN subnet.

13. TESTING & VERIFICATION

13.1 DHCP Verification




```
ipconfig
```

13.2 Intra-VLAN & Gateway Connectivity

```
ping 192.168.10.1
ping 192.168.10.11
```

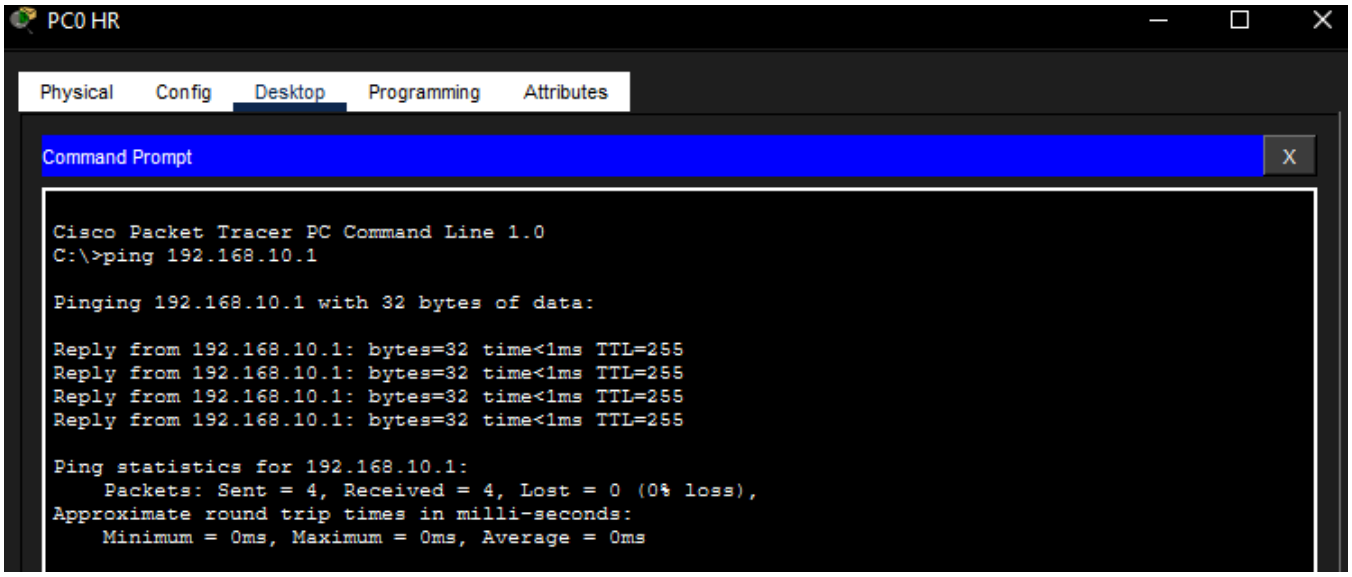
13.3 Inter-VLAN Routing

```
ping 192.168.20.10   ! from a VLAN 10 host to a VLAN 20 host
```

13.4 Internet Access (NAT)

```
ping 8.8.8.8
```

Each stage was tested only after the previous stage was confirmed working, following a bottom-up (Layer 1 → Layer 2 → Layer 3) verification approach rather than testing end-to-end connectivity first.



pinging

14. COMMON FAULTS — QUICK REFERENCE

Symptom	Likely Cause	Check
No IP / APIPA address	DHCP pool misconfigured, or trunk missing	show ip dhcp binding, show interfaces trunk
VLANs cannot reach each other	Trunk or subinterface misconfigured	show interfaces trunk, show ip interface brief
No internet access	NAT / default route missing	show ip nat translations, show ip route
EtherChannel not forming	Mismatched speed/duplex/mode	show etherchannel summary
Delay before port forwards traffic	STP listening/learning state	show spanning-tree vlan <id>

15. SECURITY AND OPERATIONAL RELEVANCE

Although implemented in a simulated environment, this project reflects concepts directly relevant to enterprise network administration and cybersecurity operations.

VLAN segmentation limits the blast radius of a compromised host by containing broadcast domains and, with further ACL work, restricting inter-department traffic. Understanding trunk behaviour, DHCP operation, and systematic fault isolation forms a foundation for later cybersecurity tasks such as network monitoring, segmentation policy, traffic analysis, and incident investigation.

16. PROJECT OUTCOME

The project successfully demonstrates the design and implementation of a segmented small-enterprise LAN containing:

- One router performing inter-VLAN routing, DHCP, and NAT
- Two access switches with redundant EtherChannel interconnection
- Three VLANs (HR, IT, Sales)
- Six end devices with dynamic addressing

- Verified inter-VLAN connectivity
- A real DHCP fault diagnosed and resolved through systematic troubleshooting

17. CONCLUSION

This lab provided practical experience in building a segmented, routed switching infrastructure using Cisco Packet Tracer, covering VLAN design, trunking, router-on-a-stick, DHCP, EtherChannel redundancy, and NAT.

The troubleshooting stage was the most instructive part of the project. Rather than assuming the DHCP pool configuration was at fault, the investigation worked outward from the router (show ip dhcp binding) to the switch (show interfaces trunk) and used CDP to confirm, rather than guess, which physical port required correction. This traced the fault to a missing trunk on the router-facing switchport — a Layer 2 issue masquerading as a DHCP problem.

The project therefore demonstrates practical understanding of VLAN segmentation, inter-VLAN routing, DHCP, EtherChannel redundancy, NAT, and methodical, layer-by-layer network troubleshooting.

18. SKILLS DEMONSTRATED

Networking

- VLAN configuration
- 802.1Q trunking
- Router-on-a-stick
- DHCP configuration
- NAT overload

Redundancy

- EtherChannel (LACP)
- Spanning Tree behaviour
- PortFast

Troubleshooting

- Interface and trunk-state analysis
- CDP-based physical topology confirmation
- DHCP-binding analysis
- Layer-by-layer fault isolation
- Ping-based connectivity testing

Tools

- Cisco Packet Tracer
- Cisco IOS CLI

19. PROJECT DESCRIPTION

Enterprise VLAN Segmentation & Inter-VLAN Routing Infrastructure

Designed and implemented a segmented small-enterprise LAN in Cisco Packet Tracer using VLANs, 802.1Q trunking, router-on-a-stick inter-VLAN routing, DHCP, EtherChannel redundancy, and NAT overload. Diagnosed and resolved a live DHCP failure by tracing the fault from the router's DHCP binding table to a missing trunk configuration on the switch, identified using CDP neighbor discovery.

FINAL PROJECT STATEMENT

Project Status: Completed / Verified

Simulation Platform: Cisco Packet Tracer

Primary Technologies: VLAN • 802.1Q Trunking • Router-on-a-Stick • DHCP • EtherChannel (LACP) • NAT Overload

Practical Level Demonstrated: Enterprise LAN Segmentation & Network Troubleshooting

MY DIFFICULTIES

- First difficulty I can face is that to configure dhcp automatically
- Second difficulty is to pinging the machines

LAB RESULT SHEET

Designing and Troubleshooting a Segmented Enterprise Network

VLANs • Router-on-a-Stick • DHCP • EtherChannel • NAT

Student Name	Ahmad Hassan	**Date**	24-09-2026
Course	Diploma in Cyber Security	**Module**	CCNA
Lab	Mega Lab 1	**Platform**	Cisco Packet Tracer
Project type	Networking Engenearing	**Overall Result**	**PASS**
Title	Multi-VLAN Enterprise Network Design with DHCP Automation, EtherChannel Redundancy & Live Fault Diagnosis		

1. OBJECTIVES ACHIEVED

#	Objective	Result
1	VLAN creation (HR, IT, Sales) on both switches	**PASS**
2	802.1Q trunk configuration (switch-to-switch, switch-to-router)	**PASS**
3	Router-on-a-stick inter-VLAN routing	**PASS**
4	DHCP pool configuration and automatic addressing	**PASS**
5	EtherChannel (LACP) redundant link bundling	**PASS**
6	NAT overload for internet access	**PASS**
7	Fault diagnosis and resolution (DHCP/trunk issue)	**PASS**

2. VERIFICATION TEST RESULTS

Test #	Command / Action	Expected Result	Result
T1	show vlan brief	VLANs 10, 20, 30 present with correct ports	**PASS**
T2	show interfaces trunk	Fa0/5 (R1 link) trunking, VLANs 10/20/30 allowed	**PASS**
T3	show etherchannel summary	Po1 status SU (S1-S2 bundle active)	**PASS**
T4	show ip interface brief (R1)	Gig0/0.10, .20, .30 all up/up	**PASS**
T5	ipconfig (PC1-PC6)	Each PC receives correct VLAN subnet IP via DHCP	**PASS**
T6	show ip dhcp binding	Active leases listed for all connected PCs	**PASS**
T7	ping <default gateway>	Reply received from VLAN gateway	**PASS**
T8	ping <PC in different VLAN>	Successful inter-VLAN reply (routing confirmed)	**PASS**
T9	show ip nat translations	Translation entries appear after outbound traffic	**PASS**
T10	ping 8.8.8.8	NAT translation created (reply depends on outside link)	**PASS**

3. FAULT LOG

Fault Observed	Root Cause	Corrective Action	Status

PCs received APIPA (169.254.x.x) address instead of DHCP lease	Router-facing switch port (Fa0/5) left in default access mode — VLAN tags dropped before reaching R1	Identified correct port via show cdp neighbors; applied switchport mode trunk + allowed vlan all; added PortFast on access ports	**PASS**

4. SKILLS ASSESSED

Skill Area			Demonstrated
VLAN segmentation & port assignment			**PASS**
802.1Q trunking configuration			**PASS**
Router-on-a-stick inter-VLAN routing			**PASS**
DHCP pool design and verification			**PASS**
EtherChannel (LACP) redundancy			**PASS**
NAT overload configuration			**PASS**
Systematic, layer-by-layer troubleshooting			**PASS**
CDP-based physical topology verification			**PASS**

5. MY DIFFICULTIES

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Student Signature: _____ Instructor Signature: _____

Final Status: COMPLETED / VERIFIED